

HW5 – EDA for heart failure dataset

1. Create “HW5.ipynb” – this will be the file for your homework.
2. In this homework you will analyze the Kaggle Heart Failure Dataset.
3. Your task is to explore this dataset. Here is an example of what you can do:
 - a. check the features and explanations of their meaning (for that you can check Data Card on Kaggle and google the topic a little bit to gain some understanding of the problem and related features),
 - b. check features effect on target variable (heart failure),
 - c. check for missing data and outliers.
 - d. bivariate/multivariate data analysis,
 - e. suggest for possible feature engineering or variable encoding,
 - f. provide some conclusions in the end about the dataset.
4. Format your IPYNB notebook as a **comprehensive report**, as you would do for a person who **is not familiar with the data** – you need to present your results clearly and tell a story about what data is inside.
5. Create “HW5” branch in your repo.
6. Upload your IPYNB file into “HW5” and create a pull request to submit your work.

Soft Deadline for this task is 02.04.2026 23.59 MSK. After this date, **half** points will be graded for the task.

Hard Deadline for this task is 09.04.2026 23.59 MSK. After this date, **no** points will be graded for the task.

This task is a kind of a creative one. The analysis and the design of plots is completely up to you. However, I will evaluate the integrity, quality, and clearness of your report. Think of your report as of a story you will be telling someone who is not familiar with the topic. Do not forget to briefly check the meaning of the features – you do not need to be a doctor for that, just keep in mind these meanings when evaluating the values. This will help you to understand how to properly deal with them.

Good luck!